

Total No. of Questions : 8]

SEAT No. :

**PB2371**

[Total No. of Pages : 5

**[6263]-221**

**B.E. (Mechanical)**

**OPERATION RESEARCH**

**(2019 Pattern) (Semester - VII) (Elective - IV) (402045D)**

*Time : 2½ Hours]*

*[Max. Marks : 70*

*Instructions to the candidates:*

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 Q.7 or Q.8.
- 2) Answers in one answer books.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

**Q1) a)** Use simplex method to solve the following LPP. **[10]**

Maximize  $Z = 2x_1 + x_2$

Subject to constraints

$$4x_1 + 3x_2 \leq 12;$$

$$4x_1 + x_2 \leq 8;$$

$$4x_1 - x_2 \leq 8;$$

$$x_1, x_2 \geq 0;$$

b) Discuss the following terms related to LPP. **[7]**

- i) Objective function
- ii) Constraint surface
- iii) Feasible and infeasible points
- iv) Optimum solution

OR

**Q2) a)** One kind of cake requires 200g of flour and 25g of fat, and another kind of cake requires 100g of flour and 50g of fat. Find the maximum number of cakes that can be made from 5 kg of flour and 1 kg of fat assuming that there is no shortage of the other ingredients, used in making the cakes. Formulate LPP and Use Graphical method to solve. **[10]**

b) Explain the limitation of LPP in solving Engineering problem. How this is used for the optimization. **[7]**

**P.T.O.**

**Q3) a)** Discuss the following related to the transportation model. [8]

- i) Feasible solution
- ii) Optimum solution
- iii) Non-degenerate basic feasible solution
- iv) Degenerate basic feasible solution

b) Find out the initial feasible solution by Vogel's Approximation Method (VAM). [10]

|             |   | Stores |    |     |    | Availability |
|-------------|---|--------|----|-----|----|--------------|
|             |   | I      | II | III | IV |              |
| Warehouse   | A | 21     | 16 | 15  | 13 | 11           |
|             | B | 17     | 18 | 14  | 23 | 13           |
|             | C | 32     | 27 | 18  | 41 | 19           |
| Requirement |   | 6      | 10 | 12  | 15 |              |

OR

**Q4) a)** In a construction site there are four cranes. Each crane must be allocated to one job. The time required to set up each crane for each job is shown in table. Find the best assignment of cranes to the jobs, so that the total time required for completing the job is minimized. [12]

|       |   | Job |    |     |    |
|-------|---|-----|----|-----|----|
|       |   | I   | II | III | IV |
| Crane | A | 5   | 7  | 11  | 6  |
|       | B | 8   | 5  | 9   | 6  |
|       | C | 4   | 7  | 10  | 7  |
|       | D | 10  | 4  | 8   | 3  |

b) Explain the significance and applications of assignment and transportation problem. [6]

- Q5) a)** A bank has decided to modernize its office. The major elements of the project are as follows.

| Activity | Description                                  | Predecessor Activity | Duration (Days) |
|----------|----------------------------------------------|----------------------|-----------------|
| A        | Design New Premises                          | ---                  | 14              |
| B        | Obtain tenders from contractors              | A                    | 4               |
| C        | Select the contractor                        | B                    | 2               |
| D        | Arrange details with selected contractor     | C                    | 1               |
| E        | Decide which equipment is to be used         | A                    | 2               |
| F        | Arrange storage of equipment                 | E                    | 3               |
| G        | Arrange disposal of other equipment          | E                    | 2               |
| H        | Order new equipment                          | E                    | 4               |
| I        | Take delivery of new equipment               | H, L                 | 3               |
| J        | Renovations take place                       | K                    | 12              |
| K        | Remove old equipment for storage or disposal | D, F, G              | 4               |
| L        | Cleaning after the contractor finished       | J                    | 2               |
| M        | Return old equipment for storage             | H, L                 | 2               |

- i) Draw an arrow diagram for this project
  - ii) Find out the critical path
  - iii) For each non-critical activity find out the total, free and independent float or Slacks. **[12]**
- b)** Explain in brief following (Any 3) : **[6]**
- i) Gradual failure
  - ii) Sudden failure
  - iii) Progressive failure
  - iv) Retrogressive failure
  - v) Random failure

OR

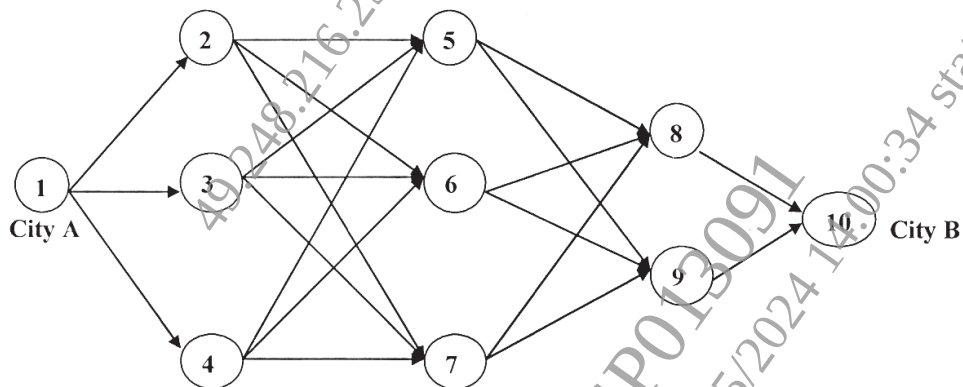
- Q6) a)** A truck owner finds from his past records that the maintenance cost per year of a truck whose purchase price is Rs. 8,000 are as follows : **[10]**

| Year                   | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    |
|------------------------|------|------|------|------|------|------|------|------|
| Maintenance Cost (Rs.) | 1000 | 1300 | 1700 | 2000 | 2900 | 3800 | 4800 | 6000 |
| Resale price (Rs.)     | 4000 | 2000 | 1200 | 600  | 500  | 400  | 400  | 400  |

Determine at which time it is profitable to replace the truck.

- b) Differentiate between CPM and PERT and also discuss various floats involved in the in CPM analysis of network. **[8]**

- Q7) a)** A salesman located in a city A decided to travel to city B. He knew the distances of alternative routes from city A to city B. He then drew a highway network map as show in following figure. The city of origin A, is city 1. The destination city B is city 10. Other cities through which the salesman will have to pass through are numbered 2 to 9. The arrow representing routes between cities and distances in kilometers are located on each route. The salesman problem is to find the shortest route that covers all the selected cities from A to B. The time for each activity is given in the table. (Solve by using Dynamic programming). **[12]**



| Activity | Duration | Activity | Duration |
|----------|----------|----------|----------|
| 1-2      | 4        | 4-5      | 6        |
| 1-3      | 6        | 4-6      | 10       |
| 1-4      | 3        | 4-7      | 5        |
| 2-5      | 7        | 5-8      | 4        |
| 2-6      | 10       | 5-9      | 8        |
| 2-7      | 5        | 6-8      | 3        |
| 3-5      | 3        | 6-9      | 7        |
| 3-6      | 8        | 7-8      | 8        |
| 3-7      | 4        | 7-9      | 4        |
|          |          | 8-10     | 7        |
|          |          | 9-10     | 9        |

- b) Explain in brief methodology of Monto Carlo Simulation. [5]

- Q8) a)** A company has five salesmen who have to be allocated to four marketing zones. The return (profit) from each zone depends upon the numbers of salesman working in that zone. The expected return for different numbers of salesman in different zones, as estimated from the past record, are given in the following table. Determine the optimum allocation policy. (Use Dynamic Programming). [12]

| Number of Salesmen | Marketing Zones |        |        |
|--------------------|-----------------|--------|--------|
|                    | Zone 1          | Zone 2 | Zone 3 |
| 0                  | 30              | 35     | 42     |
| 1                  | 45              | 45     | 54     |
| 2                  | 60              | 52     | 60     |
| 3                  | 70              | 64     | 70     |
| 4                  | 79              | 72     | 82     |
| 5                  | 90              | 82     | 95     |
| 6                  | 98              | 93     | 102    |
| 7                  | 105             | 98     | 110    |
| 8                  | 100             | 100    | 110    |
| 9                  | 90              | 100    | 110    |

- b) Explain in brief methodology involved in the simulation. [5]

